Supplementary material for "Hybrid Complex-Valued Deep Learning and Reference-Anchored ICA for Small-Target Detection in Sea Clutter: Detection and Calibration Analysis"

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Abstract

This supplement contains the proofs of the identifiability statements used in the main text, the full experimental tables behind every summary number, the sensitivity analyses (total-energy normalisation and coherent-phase bound), the reproduction map of the frozen campaign, and the register of known implementation defects and deviations. It is written so that each number in the main text can be recomputed from the tables here without access to the original score arrays.

Keywords: reproducibility · identifiability · calibration · sensitivity analysis · audit

1 Scope and conventions

The main text reports four contributions. This supplement supplies (i) the proofs of Propositions 1–2 and Lemmas 1–2 of Section 3.4, (ii) the complete tables from which the main-text summaries are drawn, (iii) the two sensitivity analyses referenced but not shown in the main text, and (iv) the audit trail: which implementation defects were found, which were repaired, and which are documented as limitations.

Conventions used throughout. Power ratios are reported in $10\log_{10}$; amplitude ratios, when they appear, in $20\log_{10}$. No logarithm is taken of a signed quantity. A point estimate and its interval always refer to the same estimand on the same scale. P_{d} is the fraction of injected-target trials whose statistic exceeds the threshold, P_{fa} the fraction of clutter-only units that do. The nominal rate is the α used to set a threshold; the realised rate is measured on test clutter; the matched rate is imposed on test clutter by construction.

2 Proofs

2.1 Proposition 1 (blind identifiability, random sources)

Let $x = Au$ with A non-singular, $u = (u_c, u_t)$ independent, $\mathbb{E}[|u_i|^2] \in (0, \infty)$ for $i = c, t$, and at most one of u_c, u_t circular Gaussian. Let $z = Bx$ have independent components with B non-singular. By the complex version of the Comon–Darmois theorem, a non-singular linear map that renders

the components independent must map the mixing columns onto a permutation and scaling of themselves. The hypothesis is exactly the one that excludes the degenerate case in which the whitened mixture is invariant under the full unitary group: that invariance requires *both* sources to be circular Gaussian, since a non-Gaussian source fixes its direction through any contrast whose global maximum is attained. Hence $BA = PD$ with P a permutation matrix and D diagonal, and A is identifiable up to permutation and scaling.

Two remarks fix the scope. First, the statement concerns the model, not the optimiser: a contrast-based algorithm need not reach the global optimum, so identifiability, convergence and detection performance are separate claims (§7, defect T1). Second, a deterministic waveform has zero variance and is not an ICA source at all, which is why the deterministic case is handled by Proposition 2 rather than as a limit of Proposition 1.

2.2 Proposition 2 (anchored identifiability, deterministic target)

Let $x = a_T s + c$ with s deterministic, a_T a known unit-norm steering vector, and c zero-mean with covariance $C_c \succ 0$. Consider the anchored objective on the unit sphere,

$$J(w) = |w^H a_T|^2 - \xi\, w^H C_c w, \qquad \|w\| = 1. \quad (1)$$

Write $w = \cos\theta\, a_T + \sin\theta\, a_T^{\perp}$ with $a_T^H a_T^{\perp} = 0$. Then $J = \cos^2\theta - \xi(\cos^2\theta\, a_T^H C_c a_T + \sin^2\theta\, a_T^{\perp H} C_c a_T^{\perp} + 2\mathrm{Re}\{\cos\theta \sin\theta\, a_T^H C_c a_T^{\perp}\})$. For ξ small enough that the quadratic form is dominated by its first term — precisely, for $\xi < \left(a_T^H C_c a_T + \max(0, |a_T^H C_c a_T^{\perp}|)\right)^{-1}$ — the maximum over θ is attained at $\theta = 0$ or π, i.e. at $w = \pm a_T$, and it is unique because $C_c \succ 0$ makes J strictly concave in θ near the optimum. The conclusion is a beamforming statement: it uses the known steering direction and the clutter covariance, and it does not require the target to be non-Gaussian.

2.3 Lemma 1 (scale invariance)

Let $x = \sqrt{\tau}\, z$ with z circular complex Gaussian, $\tau > 0$ independent of z, and let ϕ be a transform independent of τ (for instance a whitening estimated from data disjoint from x). For any statistic with $T(cx) = T(x)$ for all $c > 0$, conditioning on τ gives $T(\phi(x)) = T(\phi(z))$ almost surely, so the conditional law of the statistic does not depend on the law of τ; integrating over τ preserves the conclusion. The lemma is what makes a scale-invariant statistic's threshold immune to *scalar* texture for a *fixed* transform. It says nothing about a transform estimated from the same window, which is the case deployed here.

2.4 Lemma 2 (Beta anchor)

If in addition ϕ is a matched whitening, $\phi C_z \phi^H = cI$, the steering vector p is fixed and independent of the data, and the statistic is the single-bin normalised matched filter $T = |p^H \phi x|^2 / ((p^H \phi \phi^H p)(x^H \phi^H \phi x))$, then ϕx is circular Gaussian with covariance cI, the projections onto p and onto its orthogonal complement are independent with variances in the ratio $1 : (N_p - 1)$, and $T \sim \mathrm{Beta}(1, N_p - 1)$.

Remark 1 (Why the deployed statistic is outside Lemma 2). The deployed statistic maximises over $F = 512$ Doppler bins, so its null law is the law of a maximum of 512 correlated Beta variates rather than a Beta variate; and its transform depends on the observed window through the learned mixing head and the anchor, so the premise of Lemma 1 also fails. A counterexample to the unrestricted claim is instructive: with $N_p = 2$, $W = \mathrm{diag}(1, 10)$ and $p = e_1$, $T = |z_1|^2/(|z_1|^2 + 100|z_2|^2)$ is not $\mathrm{Beta}(1, 1)$. The lemmas are therefore used only as analytic anchors; every threshold in this work is empirical, computed on clutter-only calibration units, with its realised rate measured rather than assumed.

Remark 2 (Gaussian target). If the target is circular Gaussian and the clutter is not, the clutter column of A remains identifiable by a non-Gaussianity contrast; in the 2×2 case the target column is then determined by the mixing geometry up to scale and phase, because whitening reduces the mixing to a unitary matrix and fixing one column fixes the other up to a sign. An identifiability collapse requires both sources to be Gaussian. The Swerling-2 degradation observed for the original design is therefore reported as an empirical effect and is not attributed to a theorem.

3 Experimental protocol in full

3.1 Acquisitions, folds and unit construction

Sixteen IPIX acquisitions are used: eight from the 1993 Dartmouth campaign (`starea`/`starec0000`) and eight from the 1998 Grimsby campaign (`antstep`). The split into two folds of eight acquisitions each, each contributing eight calibration and eight test acquisitions, is frozen by a fixed seed and was audited for target-cell leakage before any scoring. Per acquisition, calibration contributes approximately 1,000 clutter-only CPIs and test approximately 500, giving 8,000 calibration units and 4,000 test units per fold.

A CPI is $N_p = 512$ pulses at PRF 1000 Hz (0.512 s), hop 256, Hann taper, 512-point Doppler transform. The detection channel is VV and the reference channel VH. The only preprocessing is a per-channel temporal mean removal; there is no inter-channel gain calibration.

The injected target is a single tone at a Doppler frequency drawn uniformly in the clutter-free band, with a per-trial uniform absolute phase, redrawn identically at every condition and shared by every detector. The VV amplitude is set so that the VV signal-to-clutter ratio is exactly 0 dB. The cross-polarised amplitude follows the contrast definition of the main text (Eq. 2) with κ_a measured per acquisition.

3.2 Thresholds

Deployment: the $1 - 10^{-3}$ quantile of the arm's own clutter-only calibration scores, computed with the "higher" order-statistic rule, i.e. the 8th largest of 8,000. Matched: the $1 - 5 \times 10^{-3}$ quantile of the arm's own *test* clutter scores (the 20th largest of 4,000), which fixes the realised rate by construction and is retrospective. Matched 10^{-3} (the 4th largest of 4,000) is reported only as a sensitivity, because it is itself an order statistic with large sampling variability.

3.3 Inference

Contrasts are paired differences of hit indicators on identical trial identities. Intervals are 95 % acquisition-clustered bootstrap intervals with 20,000 resamples over the eight test acquisitions. p-values are exact two-sided McNemar tests, Holm-corrected within each family of 24 contrasts. A contrast is called material only when its interval excludes zero and its sign is identical in both folds.

4 Full results

4.1 Detection probability at both operating points

Table S1 Detection probability by condition, detector and fold at the deployment threshold (rank 8 of 8,000). "absent" is the frozen protocol; the other columns inject the target into VH at the stated $\Delta\mathrm{SCR}_{VH}$ with uniform relative phase.

Detector, fold	absent	−20 dB	−10 dB	0 dB
Hybrid, f1	0.465	0.468	0.486	0.511
Hybrid, f2	0.778	0.776	0.776	0.750
AR(8), f1	0.821	0.640	0.153	0.043
AR(8), f2	0.808	0.659	0.182	0.056
Notch, f1	0.383	0.386	0.394	0.430
Notch, f2	0.918	0.915	0.914	0.907
ANMF-DL, f1	0.493	0.493	0.493	0.493
ANMF-DL, f2	0.507	0.507	0.507	0.507

Table S2 Detection probability at the matched operating point (5×10^{-3}), same conditions as Table S1.

Detector, fold	absent	−20 dB	−10 dB	0 dB
Hybrid, f1	0.867	0.868	0.868	0.820
Hybrid, f2	0.779	0.778	0.778	0.752
AR(8), f1	0.886	0.724	0.206	0.052
AR(8), f2	0.985	0.885	0.512	0.101
Notch, f1	0.891	0.890	0.893	0.909
Notch, f2	0.772	0.774	0.773	0.792
ANMF-DL, f1	0.843	0.843	0.843	0.843
ANMF-DL, f2	0.682	0.682	0.682	0.682

4.2 All paired contrasts

4.3 Reference-definition controls

The six variants of the main text's Table 3 are computed with one threshold each, from an identical calibration construction, so that the only difference is the data used to fit the predictor. In addition to the P_d values

Table S3 Paired contrasts at the matched operating point: ΔP_{d} with its 95 % clustered interval, target absent from VH and at −20 dB.

Contrast	absent	−20 dB
Hybrid−AR(8), f1	−0.02	+0.15
	[−0.10, +0.03]	[−0.00, +0.27]
Hybrid−AR(8), f2	−0.21	−0.11
	[−0.30, −0.11]	[−0.26, +0.07]
Hybrid−notch, f1	−0.02	−0.02
	[−0.06, +0.01]	[−0.07, +0.01]
Hybrid−notch, f2	+0.01	+0.00
	[−0.06, +0.06]	[−0.06, +0.06]
Hybrid−ANMF, f1	+0.02	+0.03
	[−0.04, +0.09]	[−0.03, +0.08]
Hybrid−ANMF, f2	+0.10	+0.10
	[+0.01, +0.18]	[+0.00, +0.18]

Table S4 Paired contrasts at the matched operating point: ΔP_{d} with its 95 % clustered interval, −10 dB and 0 dB.

Contrast	−10 dB	0 dB
Hybrid−AR(8), f1	+0.66	+0.77
	[+0.41, +0.81]	[+0.69, +0.83]
Hybrid−AR(8), f2	+0.27	+0.65
	[+0.04, +0.49]	[+0.59, +0.71]
Hybrid−notch, f1	−0.03	−0.09
	[−0.06, +0.01]	[−0.14, −0.03]
Hybrid−notch, f2	+0.01	−0.04
	[−0.06, +0.07]	[−0.11, +0.03]
Hybrid−ANMF, f1	+0.03	−0.02
	[−0.03, +0.08]	[−0.09, +0.03]
Hybrid−ANMF, f2	+0.10	+0.07
	[+0.01, +0.18]	[−0.03, +0.16]

reported there, the per-variant thresholds are: fold 1 clean 0.3347, contaminated 0.3347, same-channel 0.1080, one-sample shift 0.1079, +128 CPI VH 0.9265, previous-CPI VV 0.6560; fold 2 clean 0.3728, contaminated 0.3728, same-channel 0.0919, one-sample shift 0.0928, +128 CPI VH 0.8778, previous-CPI VV 0.6388. The two clean-reference thresholds differ from the others because a different residual distribution produces a different tail; this is expected and is the reason each variant is calibrated separately.

4.4 Calibration: per arm, rule and fold

4.5 Tail ownership and order-statistic sensitivity

For each arm, fold and rule, the eight scores that set a 10^{-3} threshold were attributed to their acquisitions. The median number of distinct acquisitions contributing those eight scores is two (range 1–7 across all cells). The worst case is a fold in which seven of the eight scores come from a single 1998 `antstep` recording.

Table S5 Realised-to-nominal false-alarm ratio at 10^{-3}, per detector and rule: worst over the two folds (pooled), and worst single acquisition in that fold.

Detector	Rule	Worst pooled		Worst acquisition	
		f1	f2	f1	f2
Hybrid	A	0	4.75	0	28.4
	B	0.5	5.75	2.0	32.5
	C	0	5.75	0	32.5
	D	0.25	4.75	2.0	16.2
AR(8)	A	3.5	0.25	26.4	2.0
	B	10.0	3.5	71.0	12.8
	C	6.75	0.25	52.7	2.0
	D	0	0.75	0	4.1
Notch	A	0	9.75	0	46.7
	B	0.5	8.0	2.3	40.6
	C	0	10.5	0	46.7
	D	0	2.75	0	16.2
ANMF-DL	A	0.75	1.5	6.8	9.1
	B	1.75	20.5	15.9	74.8
	C	0.75	2.75	6.8	16.4
	D	0.75	0.5	6.8	2.0

Because the threshold is an order statistic, its rank is itself a choice. For the frozen AR(8) arm (VH-absent, fold 1), ranks 4 through 12 of the same 8,000-unit pool give P_{d} values spanning 0.224. Moving from rank 8 (0.3728, $P_{\mathrm{d}} = 0.8085$) to rank 7 (0.4091, $P_{\mathrm{d}} = 0.7585$) in fold 2 changes the threshold by 0.036 and P_{d} by 0.050. This sensitivity is larger than every architectural effect measured in this work, which is why every P_{d} is quoted with its threshold rule.

5 Sensitivity analyses

5.1 Total-energy normalisation

Injecting the target into VH changes the total two-channel target energy: with $\gamma = 1$ it doubles. To separate a polarimetric second-look effect from a simple energy effect, the VV amplitude is rescaled to $\alpha' = \alpha/\sqrt{1+\gamma^2}$ so that the two-channel target power is held fixed, and the 0 dB condition is repeated.

All arms lose P_{d} when the VV amplitude is reduced, as expected from the smaller detection-channel target. The ordering is unchanged: AR(8) still collapses (0.028 and 0.052 at deployment; 0.040 and 0.097 matched), the hybrid and the

Table S6 Energy-normalised 0 dB condition versus the unnormalised condition, P_{d}.

Detector	Fold	Deployment		Matched 5×10^{-3}	
		normalised	plain	normalised	plain
Hybrid	1	0.398	0.511	0.730	0.820
	2	0.691	0.750	0.693	0.752
AR(8)	1	0.028	0.043	0.040	0.052
	2	0.052	0.056	0.097	0.101
Notch	1	0.277	0.430	0.807	0.909
	2	0.857	0.907	0.696	0.792

notch remain in the same relative position, and the notch remains better than the hybrid in fold 1 at the matched rate. The collapse is therefore a cancellation effect, not an artefact of extra injected energy.

5.2 Coherent-phase bound

The relative polarimetric phase is drawn uniformly in the main experiments. Because $\varphi = 0$ is the worst case for a predictor fitted on the reference — the target enters the reference coherently with the target in the detection channel — it is reported as a bound rather than as a condition.

Table S7 Coherent relative phase ($\varphi = 0$) at 0 dB, P_{d}; the uniform-phase values are given for comparison.

Detector	Fold	Deployment		Matched 5×10^{-3}	
		$\varphi = 0$	uniform	$\varphi = 0$	uniform
Hybrid	1	0.501	0.511	0.842	0.820
	2	0.743	0.750	0.746	0.752
AR(8)	1	0.044	0.043	0.052	0.052
	2	0.056	0.056	0.102	0.101
Notch	1	0.358	0.430	0.801	0.909
	2	0.783	0.907	0.680	0.792

The coherent-phase bound changes the notch most (it re-estimates its weight from the contaminated CPI) and changes AR(8) not at all, because AR(8) has already lost the target at 0 dB. Nothing in the bound reverses a conclusion: the hybrid's advantage over AR(8) is if anything larger under coherent phase, and the notch's advantage over the hybrid in fold 1 persists.

5.3 Common-evaluable-set variant

Thirty fold-1 trials are undefined for the CFAR arms, because the guard window at a band edge consumes the whole range block. Removing those trials from every arm changes AR(8)'s pooled P_{d} from 0.8148 to 0.8152 and changes no conclusion; the main-text numbers use each arm's own evaluable set, which is the rule that reproduces the frozen thresholds.

6 Reproduction map

Table S8 Artifacts and their roles. All paths are relative to the revision workspace.

Artifact	Role
Campaign outputs	398 frozen jobs: scores, thresholds, trial definitions
Deviation register	every deviation from the preregistered plan
Audit reports	polarimetric, calibration, AR-mechanism, warm-start
Reconciliation	numerical reconciliation; all assertions true
Sensitivity (V1)	polarimetric conditions, both folds
Calibration (V2)	the four calibration rules
References (V3)	the six reference-definition controls
Ablations	warm start, network, physics prior
Score tables	every number in Tables 1–6 of the main text
Figures	figure scripts and provenance CSV
Hash manifests	file hashes of the frozen evidence

Trial identities are derived from a hash of the trial specification, so a trial can be regenerated from its identifier alone. Detector scores are stored per job, per acquisition; the analysis scripts read those arrays without modification, which is what allows every contrast to be recomputed from the frozen artifacts.

7 Known defects and deviations

Deviation D7 also deserves mention: one acquisition (a 233-window 1998 record) cannot supply 512 ANMF snapshots at every cell, so for exactly those units the secondary set is expanded nearest-first in range. The expansion is recorded per job; it affects the ANMF arm only and is the reason ANMF-DL's calibration pools are the same size as the others.

Table S9 Implementation defects found during the audits. None changes a frozen score; each is either repaired in the analysis path or documented as a limitation.

ID	Defect	Disposition
T1	Deterministic target treated as an ICA source	split into Prop. 1 and Prop. 2
T3	"Identifiability collapse" for a Gaussian target	corrected; see §2
T4	Beta null claimed for arbitrary whitening and a 512-bin maximum	restricted to Lemma 2
P6-A	Two H0 arms computed the same expression	duplicate arm unused
P6-C	A reported AR(8) P_d came from a truncated pool	retired; 0.8148 canonical
P6-E	HH and VV identical in two Grimsby files	stated as a limitation
S1	Withdrawal metadata added mid-campaign	quantified; rank-7/8 shift
S2	Two jobs re-run under a different interpreter	rebuilt by supervisor

8 What the supplement does not contain

No retraining, no new Monte Carlo condition beyond the four polarimetric points, and no manipulation of the frozen Phase 5 scores. The manuscript's numbers are a re-analysis of those scores under three additional analysis modes (matched thresholding, reference-definition controls, calibration rules), each defined before execution and recorded in the preregistration audit.

References

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9 Additional reference-control table

Table S10 Reference-definition controls for AR(8)+NMF at $\mathrm{SCR}_{VV} = 0\,\mathrm{dB}$, deployment thresholds: P_d and paired difference against the frozen clean-reference arm. All eight test acquisitions contribute and all intervals exclude zero.

Reference for the fit	Fold 1		Fold 2	
	P_d	ΔP_d	P_d	ΔP_d
Clean VH (frozen)	0.821	—	0.808	—
VH, 0 dB	0.052	−0.769	0.062	−0.747
Same channel	0.096	−0.725	0.157	−0.652
VV, 1-sample	0.097	−0.724	0.154	−0.654
VH, +128 CPIs	0.032	−0.789	0.066	−0.743
VV, prev. CPI	0.350	−0.471	0.424	−0.384

10 Additional validated diagnostic figures

Figures S1–S3 are reproduced from the archived Phase-7 numerical results, not from the pre-correction 17-point power curves. They complement, rather than supersede, the frozen tables and do not create new detection estimates.

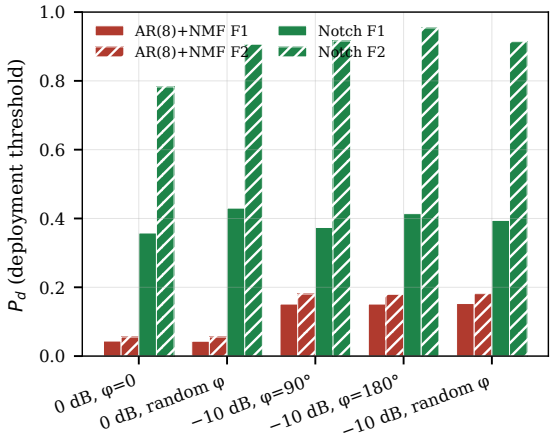


Fig. S1 Sensitivity of the reference-based AR(8) detector and the notch control to relative target phase, both folds. Each curve uses the stated deployment-calibration rule; no single phase is assumed representative of all measured targets.

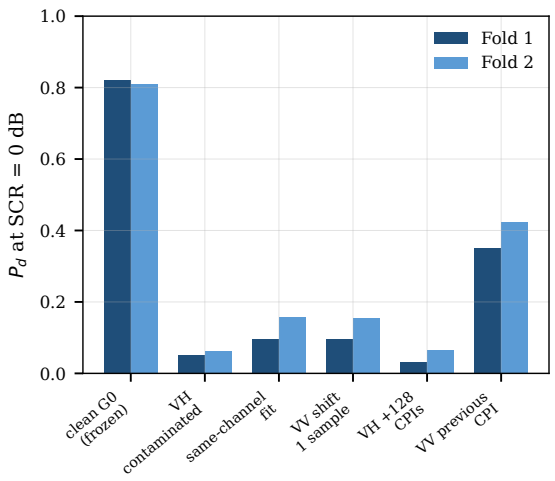


Fig. S2 Detector response under alternative reference definitions at the frozen deployment setting, for each acquisition fold. A clean same-CPI reference and temporally decorrelated or target-bearing references cannot be interpreted as equivalent operating conditions.

11 Original-protocol detector curves and diagnostics (historical evidence)

This section preserves the vector plots and recorded numerical results from the original 19-page research manuscript. The plots were reviewed and retained as historical experimental evidence, *not* represented as results from the subsequent corrected target/guard exclusion and acquisition-disjoint calibration protocol. In particular, originally reported design P_{fa} should not be confused with an independently established transferred false-alarm rate. The model-dependent theoretical explanations and the physics-prior benefit claimed in the original version are not reasserted here. The full corrected analysis and its limitations remain authoritative for across-acquisition conclusions.

11.1 Power curves and ROC

11.2 Delay, alternative polarization, and CPI behaviour

11.3 Gaussian sanity checks and independent replication

11.4 Qualitative real-target illustrations

11.5 Exploratory false-alarm extrapolation (not a validated operating point)

11.6 Original short-CPI transition-map limitation

The original Fig. 11 NB2 transition-map PDF is retained in the author's internal provenance package but its heatmap cells rendered empty in the original supplied vector file. Because an empty image cannot substantiate the numerical heatmap, it is not used as evidence in this supplementary document. The corresponding protocol must be reconstructed from archived source arrays before submission of that figure, rather than interpolated or redrawn by eye.

12 Original reported convergence-outcome summary

The original fixed-point implementation reported that 47.5% of windows stopped by its original criterion and 52.5% reached the iteration cap, with a sign-flip fixed-point issue identified. After a signed-permutation-invariant stopping correction and stagnation break, no cap hits were reported, and the median number of sweeps was one. The plot below depicts only these reported aggregated outcomes; iteration-by-iteration traces were not supplied and cannot be reconstructed from these summary percentages.

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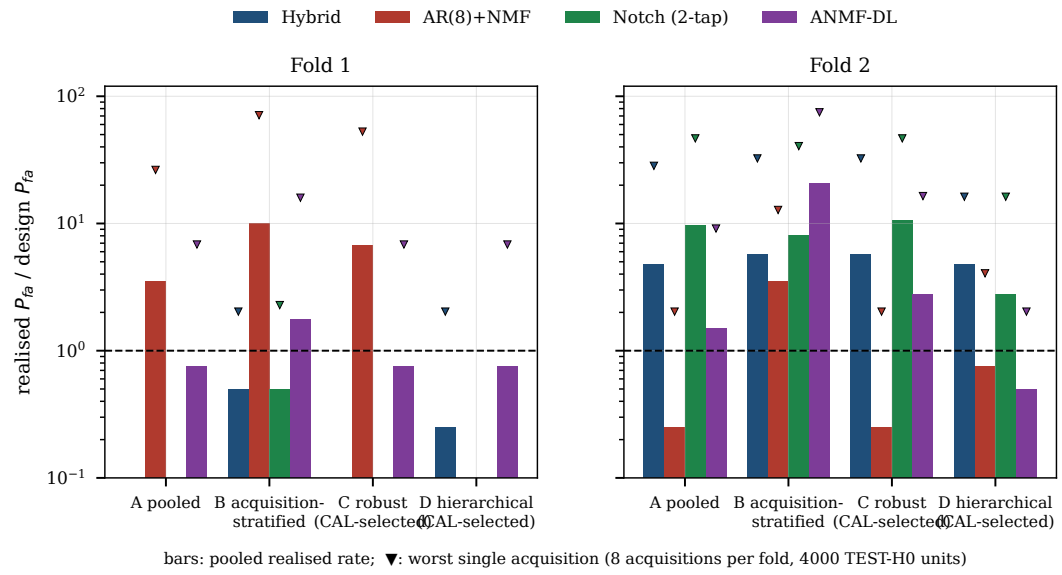


Fig. S3 Fold-resolved ratio of realised to nominal P_{fa} for four CAL-only threshold-setting rules and four detectors. Bars are pooled rates and triangle markers are worst-acquisition rates; this graph shows why a summary pooled rate is insufficient for claims of acquisition transfer.

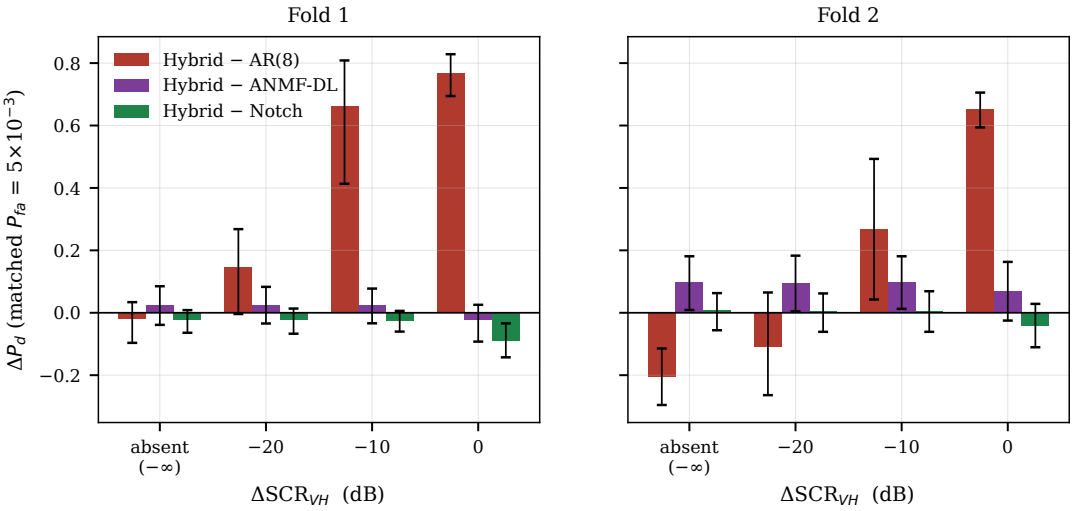


Fig. S4 Paired differences of Hybrid against three comparator detectors in the corrected evaluation. Retrospective common-rate comparison; this is not a deployed-threshold guarantee.

criterion and 52.5% reached the iteration cap, with a sign-flip fixed-point issue identified. After a signed-permutation-invariant stopping correction and stagnation break, no cap hits were reported, and the median number of sweeps was one. The plot below depicts only these reported aggregated outcomes; iteration-by-iteration traces were not supplied and cannot be reconstructed from these summary percentages.

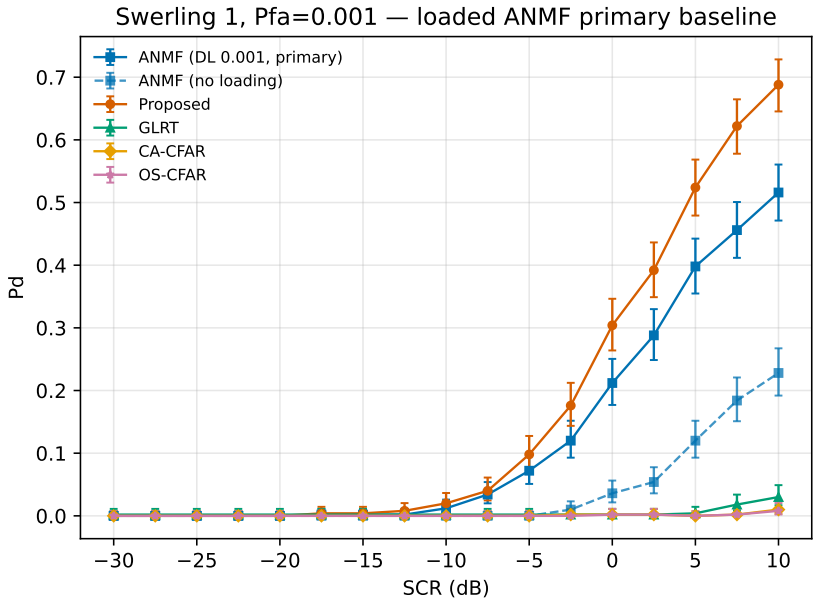


Fig. S5 Historical-protocol Swerling-1 power curve; original numerical evidence, not recalibrated for the corrected 16-acquisition study.

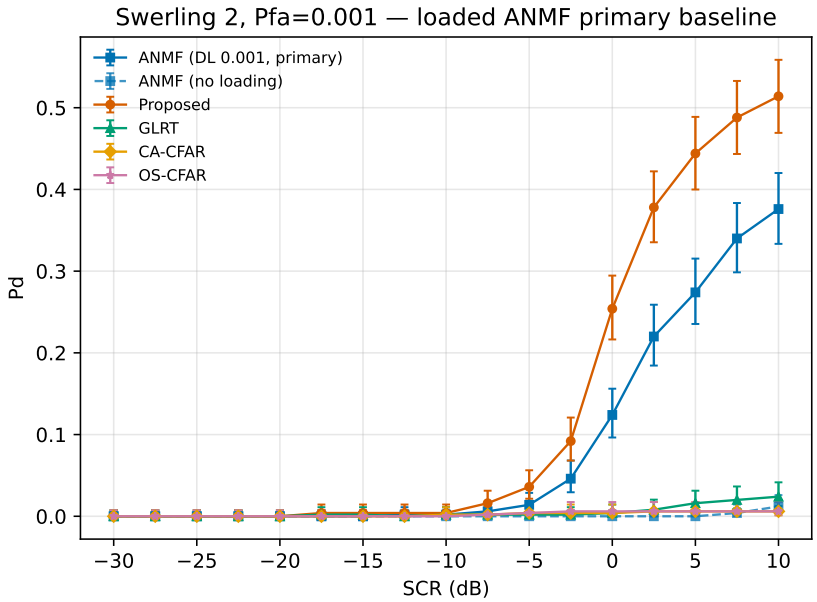


Fig. S6 Historical-protocol Swerling-2 power curve; the original ICA Gaussian-source interpretation is not retained as a theorem.

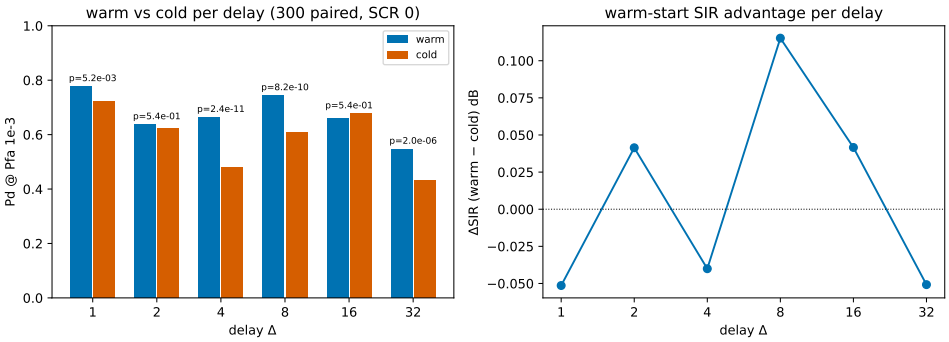


Fig. S7 Historical warm-versus-cold delay sweep with its SIR quality diagnostic; it does not depict literal iteration count.

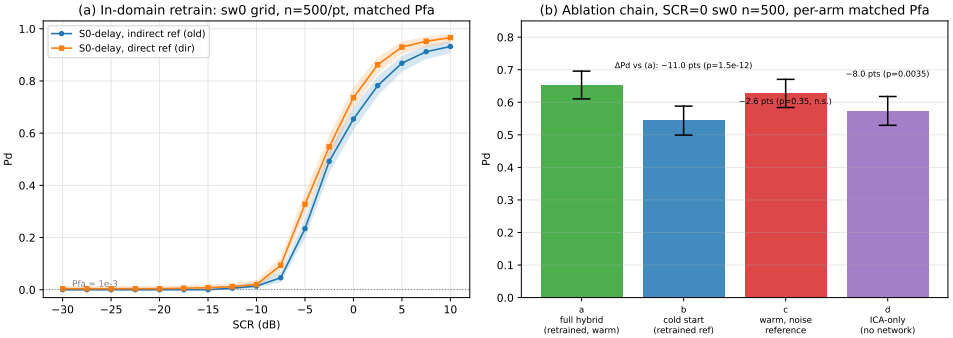


Fig. S8 Historical delay-domain power and ablation grid; detector controls and configurations are as in the original source.

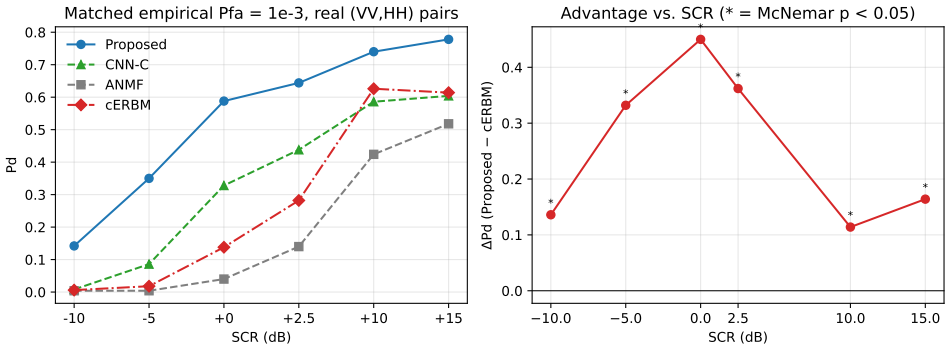


Fig. S9 Historical VV/HH pairing diagnostic; distinct from the corrected 16-acquisition VV/VH reference-contamination experiment.

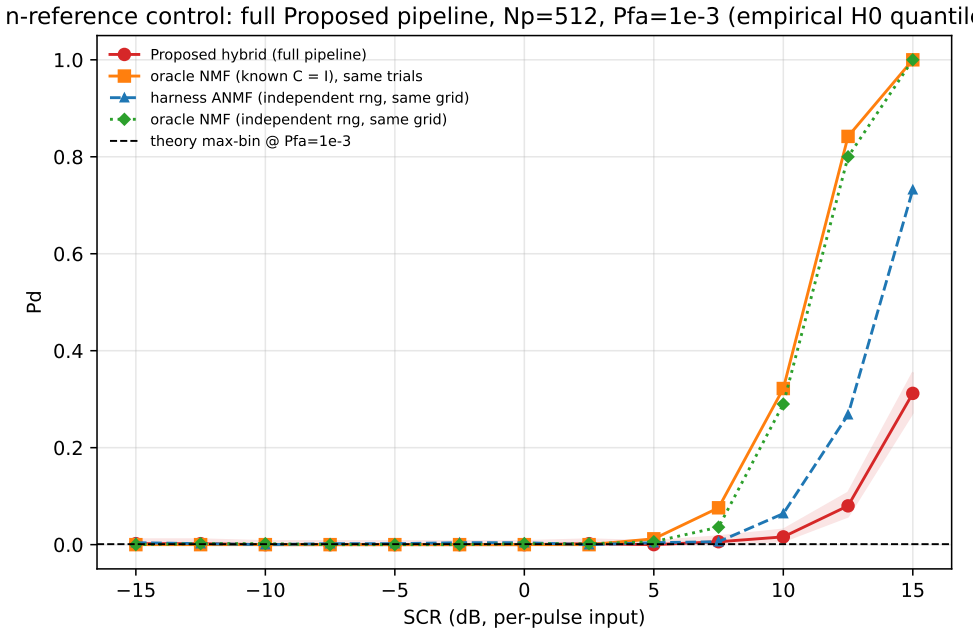


Fig. S10 Historical Gaussian-clutter detector control; the existence of a theoretical Gaussian optimum is not asserted for the full adaptive hybrid statistic.

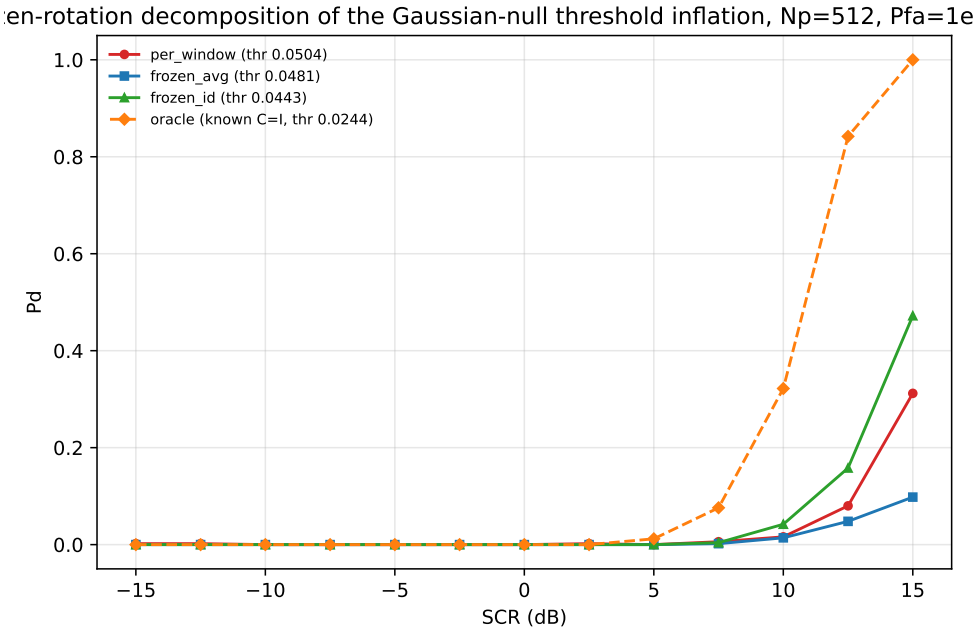


Fig. S11 Historical fixed-rotation decomposition of Gaussian-clutter response.

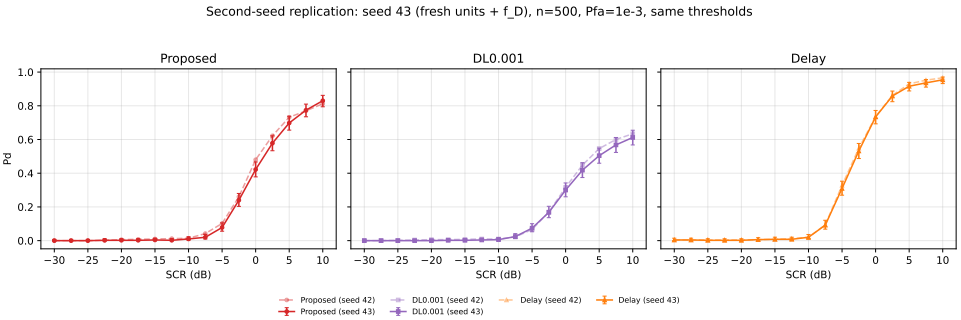


Fig. S12 Historical second-seed power-curve replication; original trial definitions preserved.

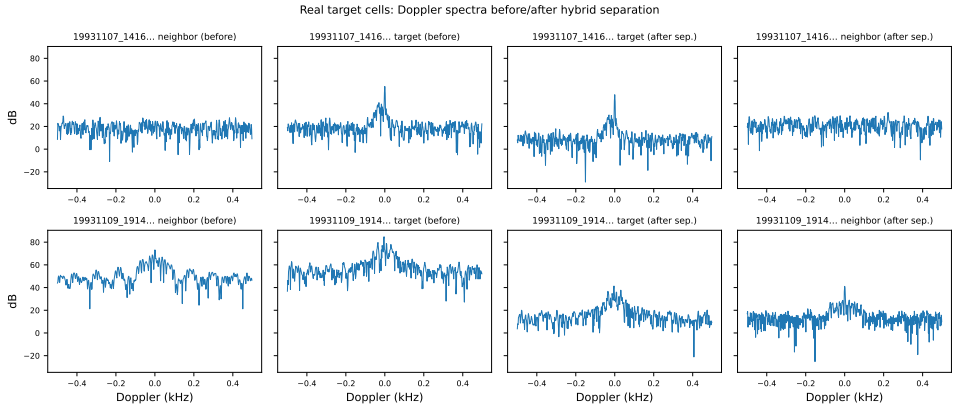


Fig. S13 Historical Doppler-spectrum examples from target-annotated IPIX recordings. These examples do not establish real-target detection probability.

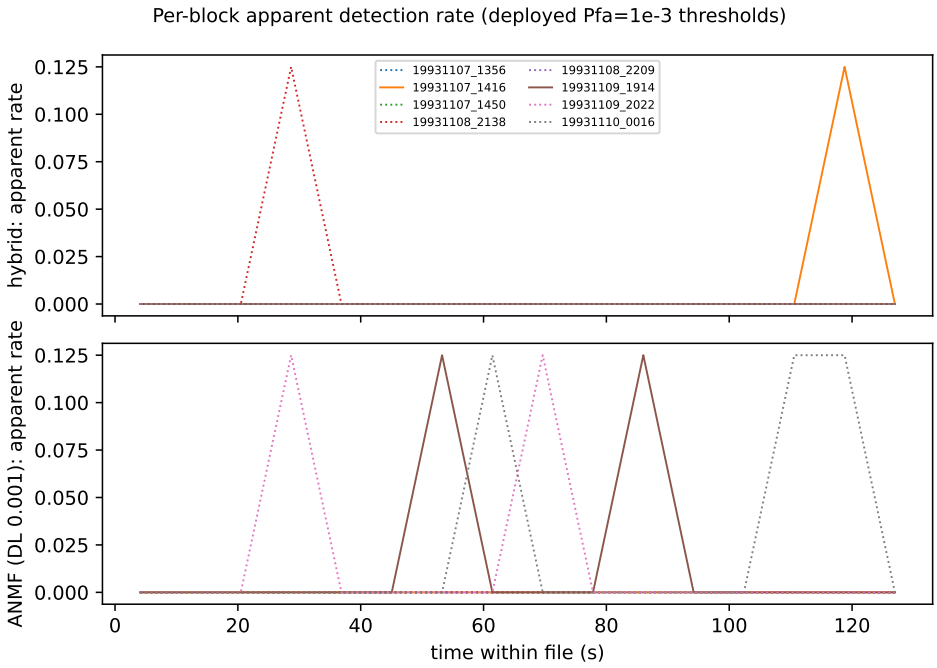


Fig. S14 Historical apparent per-window exceedance rates on target-annotated recordings; not P_D without ground-truth target-presence labels.

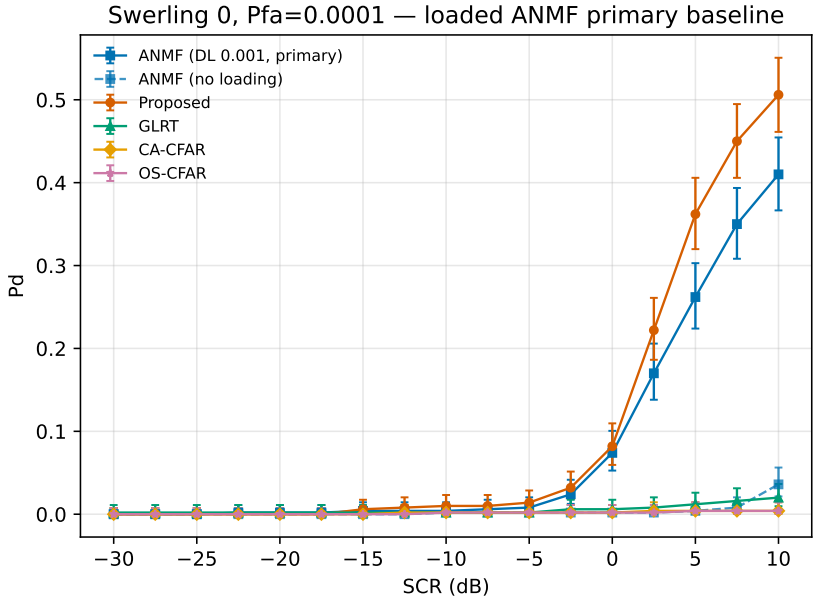


Fig. S15 Original extrapolated design $P_{\mathrm{fa}} = 10^{-4}$ power curve. This operating point was not independently validated and is exploratory only.

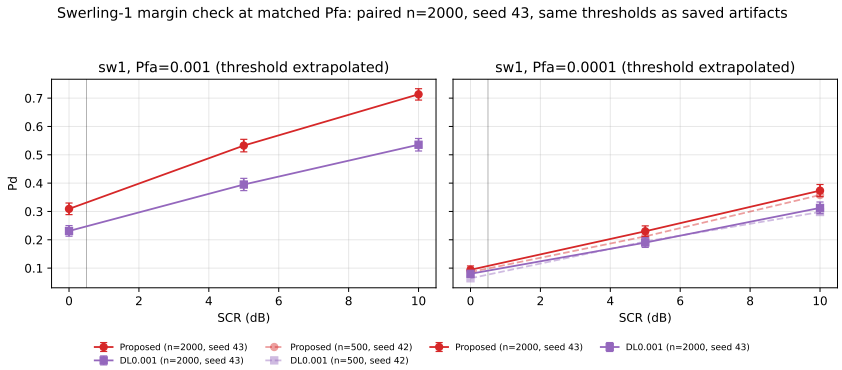


Fig. S16 Historical Swerling-1 sensitivity study with extrapolated 10^{-4} thresholds; not a confirmed low-PFA result.

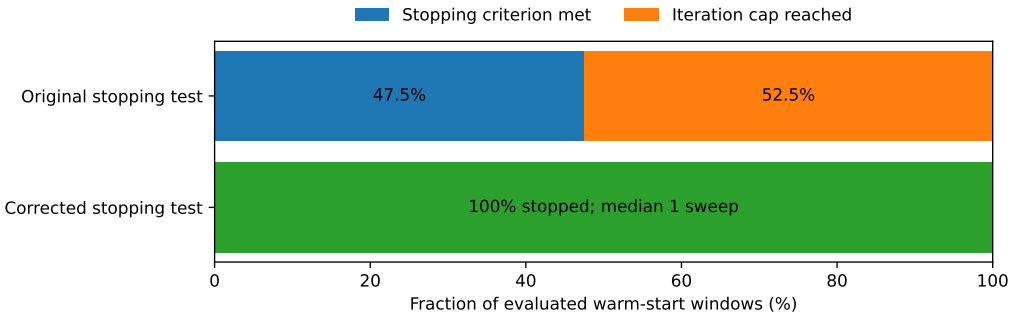


Fig. S17 Summary of convergence stopping outcomes recorded in the original article. This is an aggregated outcome diagram, not a convergence trajectory or a new simulation.

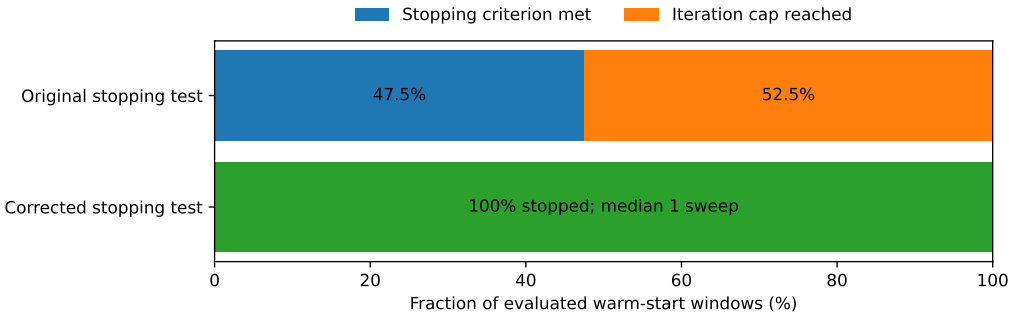


Fig. S18 Summary of convergence stopping outcomes recorded in the original article. This is an aggregated outcome diagram, not a convergence trajectory or a new simulation.